

CuFact

Custom Manufacturing Enterprise Platform

Project Definition & Progressive Build Plan

Nicolás Gómez Aragón

September 2026, Bogotá, Colombia

1. Project Vision

CuFact is an enterprise platform for custom plastic manufacturing. It will begin as a focused ERP-like system centered on order management and progressively expand into quoting, pricing, production, data, machine learning, automation, and intelligent agents.

The platform is intentionally designed to grow. We will not attempt to build a complete enterprise system upfront. Each stage will introduce a new capability only when the previous layer is functional and when the added complexity provides a meaningful business or engineering challenge.

2. Domain

Initial domain: custom plastic manufacturing / Engineer-to-Order (ETO). The company produces plastic parts or products according to customer requirements rather than relying exclusively on standard catalog items.

The domain is broad enough to support industrial manufacturing concepts while remaining accessible for experimentation. 3D printing can also provide a practical way to prototype selected physical concepts without requiring industrial manufacturing infrastructure.

3. Product Positioning

CUFACCT should initially be treated as an ERP specialized in custom manufacturing, while keeping the architecture open enough to evolve into a broader enterprise platform.

- Initial product: CUFACT ERP
- Longer-term vision: CUFACT Platform
- Core domain: Custom / Engineer-to-Order Manufacturing

4. Architectural Philosophy

Salesforce and AWS are not separate applications. They are two layers of the same platform.

Layer	Primary Role	Examples
Salesforce	User-facing enterprise application	Users, CRM, configuration, orders, workflows, visibility

Integration	Bidirectional communication	APIs, events, synchronization, validation
AWS	Operational backend	Services, persistence, processing, pricing, production workloads
SageMaker	Machine learning	Prediction, recommendations, model inference
Agentforce	Intelligent interaction	Assistance, explanations, controlled actions

Salesforce will be the primary interface through which enterprise users interact with CUFACT. AWS will progressively become the operational backend where data and processing required for production are handled. AWS sends relevant status and results back to Salesforce so users maintain a coherent view of the business.

5. Initial Scope

The first implementation target is deliberately narrow: Order Management.

- Customers / Accounts
- Products
- Orders
- Order Items
- Basic product configuration
- Order lifecycle and status
- Salesforce-to-AWS synchronization

Pricing, advanced production, machine learning, Agentforce, resilience, and other enterprise capabilities are deliberately deferred until later stages.

6. Initial Business Flow

The first meaningful end-to-end flow will be:

Customer → Order → Order Items → Product / Configuration → Production → Status Updates

The exact business process will evolve as we learn more about custom plastic manufacturing. The goal is to avoid locking the project into assumptions that have not yet been validated.

7. Progressive Development Stages

1. Salesforce Foundation

Build the initial enterprise data model and user-facing application in Salesforce. Establish the core entities and relationships required to support customers, products, orders, and the initial business flow.

2. Configuration & Quoting

Introduce configurable products and quoting capabilities. Salesforce handles the user experience and initial commercial logic while keeping the model extensible for future pricing complexity.

3. AWS Core Backend

Build the operational backend in AWS. Establish the services and persistence needed to receive information from Salesforce and process data that will eventually support production.

4. Salesforce ↔ AWS Integration

Create reliable bidirectional communication. Salesforce sends business information to AWS, while AWS returns processing results, operational states, and updates that users can see in Salesforce.

5. Python Pricing Engine

Introduce a dedicated pricing engine in Python, initially hosted on EC2. Pricing calculations become an external service with explicit API contracts and progressively more sophisticated rules.

6. Production Operations

Extend the backend to represent post-order operational processing, production states, manufacturing information, and synchronization of operational updates back to Salesforce.

7. Data Platform & Analytics

Centralize historical orders, configurations, costs, pricing, and production data in AWS. Build the foundation for analytics and future machine-learning workloads.

8. SageMaker & Predictive Intelligence

Use accumulated data to train and deploy predictive models. Possible applications include cost estimation, price recommendations, margin prediction, or other manufacturing forecasts.

9. Automation & Agentforce

Introduce more advanced automation and Agentforce. Agents may retrieve information, explain decisions, identify missing information, or execute tightly controlled actions through defined tools and permissions.

10. Enterprise Architecture

Harden the platform for enterprise characteristics: security, observability, resilience, scalability, performance, backups, disaster recovery, and potentially multi-region operation.

8. Guiding Development Method

The project will be developed in short, flexible sprints. The roadmap is a direction, not a fixed contract. Stages may be reordered, expanded, reduced, or replaced when the system exposes a better problem or architectural opportunity.

1. Build a small functional capability.
2. Observe the business and technical problems it creates.
3. Propose an architectural solution.
4. Implement it.
5. Test and review the result.
6. Introduce complexity only when justified.

9. Architectural Learning Objectives

- Design a strong enterprise data model.
- Understand system boundaries and ownership of data.

- Design bidirectional Salesforce/AWS integrations.
- Build and consume APIs.
- Develop production-oriented Python services.
- Handle synchronous and asynchronous processing.
- Design for failures, retries, idempotency, security, and observability.
- Build practical data pipelines.
- Integrate machine learning into an existing operational system.
- Design controlled AI/Agentforce interactions.
- Reason about scalability, resilience, and architectural trade-offs.

10. Project Rules

- **Problem before technology.**
- Functional progress before theoretical perfection.
- No technology is added merely to showcase it.
- Salesforce and AWS remain part of one coherent platform.
- Architecture decisions must have a reason and an explicit trade-off.
- The design is allowed to change as we learn.
- Each sprint should leave the system in a usable state.

11. Current Starting Point

CUFACT is ready to begin with Order Management. The immediate task is to define the initial Salesforce data model for orders and the minimum business flow required to create, process, and track an order.

Next milestone: Design the first CUFACT data model and Order lifecycle.

12. Long-Term Vision

The long-term objective is not simply to reproduce an ERP. CUFACT should become a coherent enterprise platform in which Salesforce provides the primary business interface and AWS provides the operational and computational foundation, with data, ML, automation, and intelligent agents introduced as the system's needs evolve.